complicated geometric implication for the 2-D case, which gets even more complicated for hyperspheres in higher dimensions).

- Getting right the thresholds passed to canopy clustering is complicated: because of the spherical shape of canopies, we will have to use a larger-than-needed T_1 radius, and a significant overlapping between canopies, to capture the relations between clusters in different canopies;
- When the overlapping between canopies is large, points get assigned to several canopies, and thus they will need to be processed several times for each pair of canopies: this easily becomes a computational nightmare, making vain all the effort we did to parallelize.

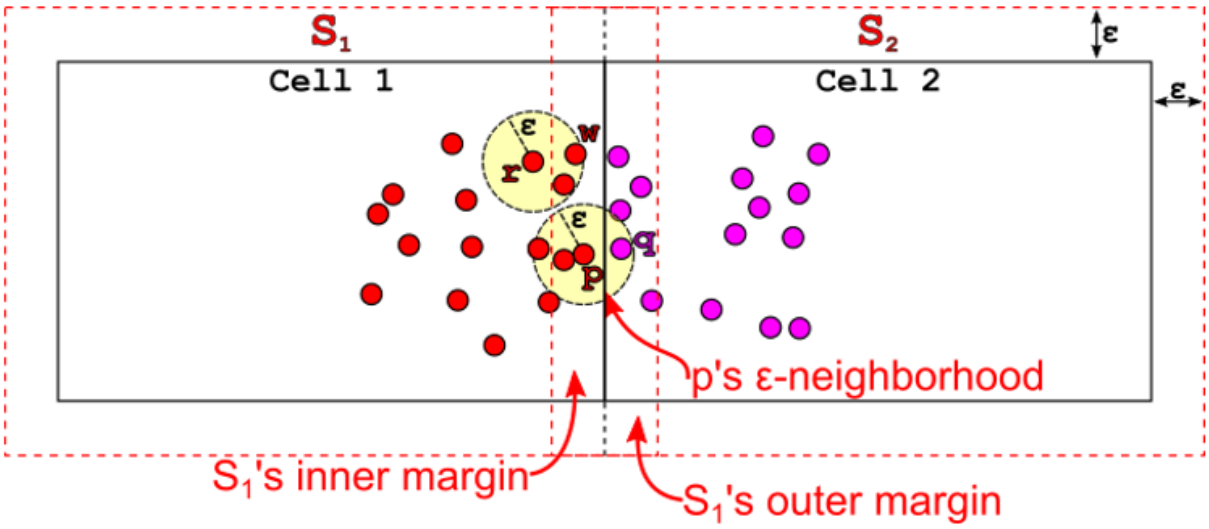


Figure 13.12 Sharding data for MapReduce DBSCAN (MR-DBSCAN). After dividing the domain in equally-sized cells, for each shard we need to include all points in one cell plus all the points within a distance equal to ε from its borders: in practice instead of the